

Supplementary information

Annual 30 m leaf-on daytime land surface temperature composites across China from 2000 to 2024

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Six macroregions used for stratification and reporting



Figure S1: Six macroregions used for stratified sampling and regional reporting. The boundaries aggregate 46 ecological regions from the China eco-geographical regionalization: TP, Tibetan Plateau; NEC, Northeast China; NCLP, North China–Loess Plateau; ANW, Arid Northwest; HSE, Humid Southeast; and SWM, Southwest Mountains. These abbreviations are used consistently in the main text and figures.

Spatial support of the 2000-2024 Landsat validation sample

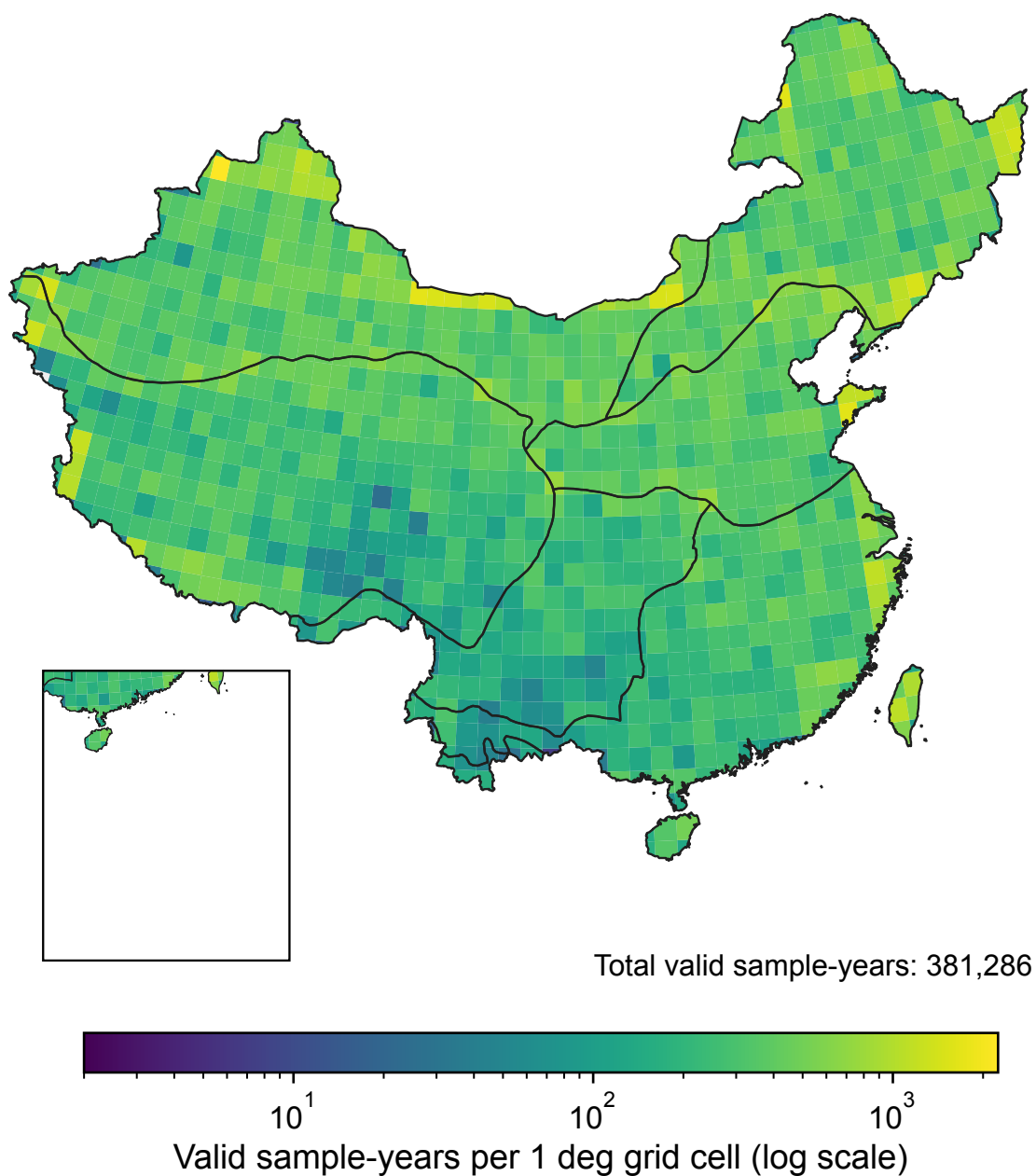


Figure S2: Spatial support of the supplementary nationwide multisensor Landsat comparison, 2000–2024. This extended archive combines Landsat 5, 7, 8 and 9, whereas the main-text comparison uses Landsat 8 alone during 2013–2024. Valid sample-years are aggregated to 1° grid cells. A sample-year requires strict cloud, shadow, snow and radiometric-saturation screening, at least three valid Landsat observations, available 3×3 ChinaTherm30 and RF-LST baseline means, and a water fraction no greater than 10%. Macroregion boundaries follow Figure S1.

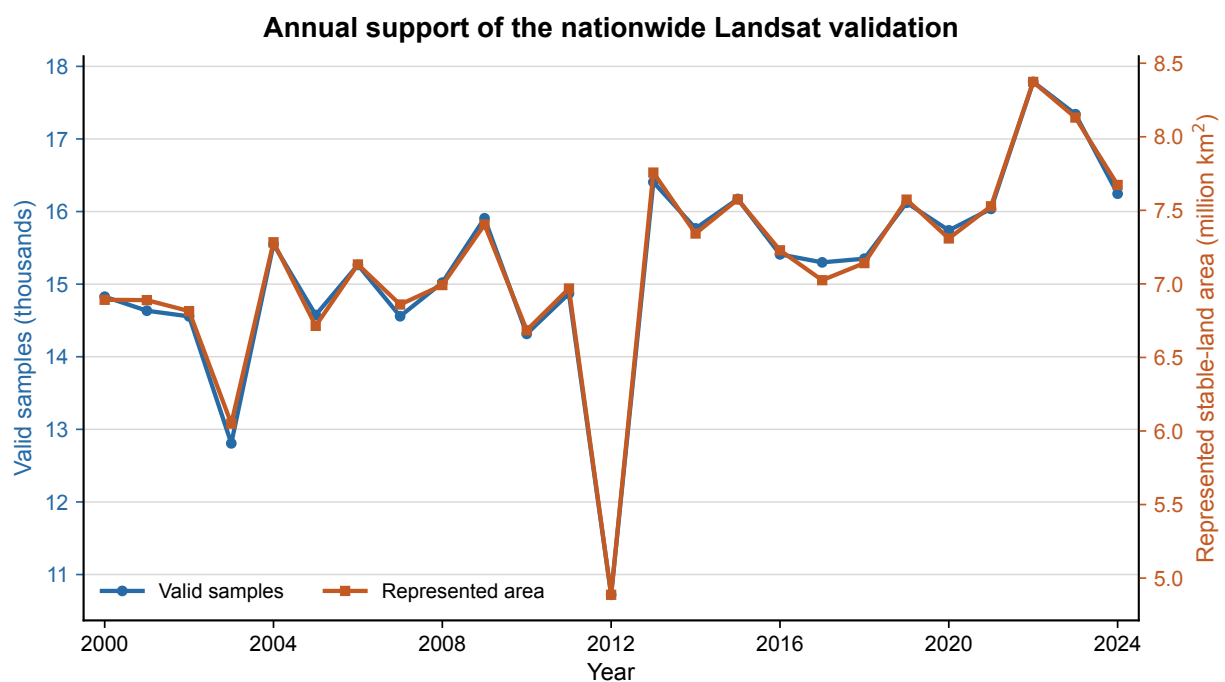


Figure S3: Annual support of the supplementary nationwide multisensor Landsat comparison, 2000–2024. Valid sample counts and represented land area correspond to the Landsat 5/7/8/9 archive in Figure S2. Counts reflect at least three quality-screened observations per sample-year, while area totals use the original stratified-sampling weights. The Landsat 8-only comparison in the main text uses its own annual valid samples during 2013–2024.

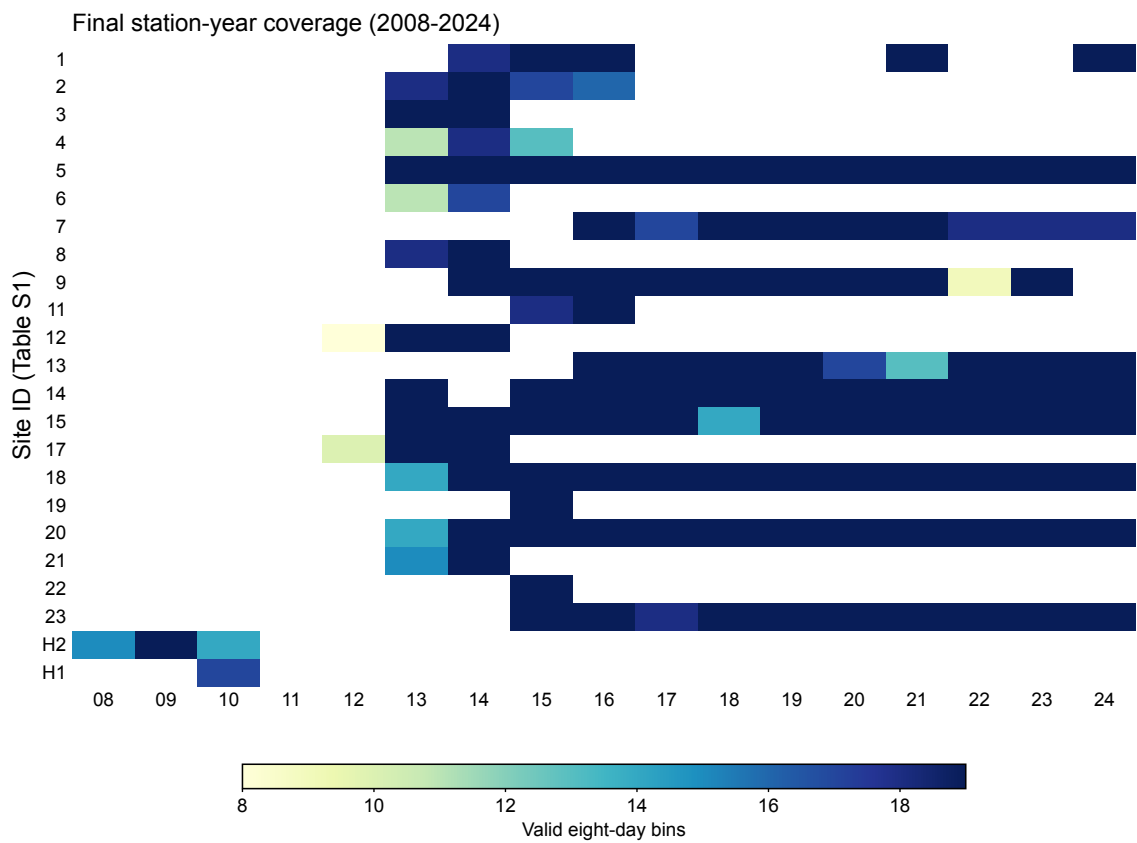


Figure S4: Temporal coverage of the final ground-validation set. Colours indicate the number of valid MOD11A2-aligned eight-day bins contributing to each station-year annual IRT composite. The final set contains 131 station-years across all 23 sites and 16 calendar years. Site IDs follow Table S1.

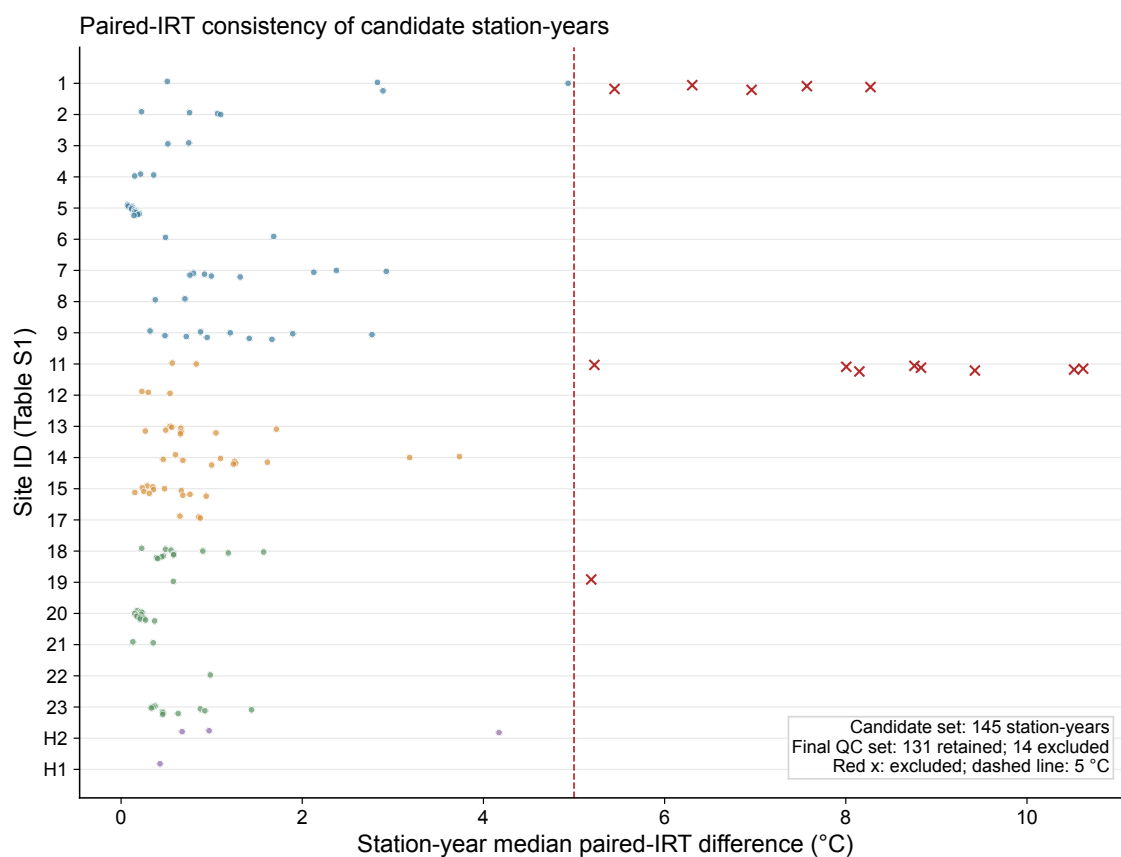


Figure S5: Paired-IRT consistency within the 145 candidate station-years. A paired-sensor diagnostic was available for 143 records. Fourteen station-years with a median daily paired-IRT absolute difference greater than 5 °C were excluded from the final ground-reference set; the remaining 131 station-years retained all 23 sites. The criterion uses only ground-sensor measurements.

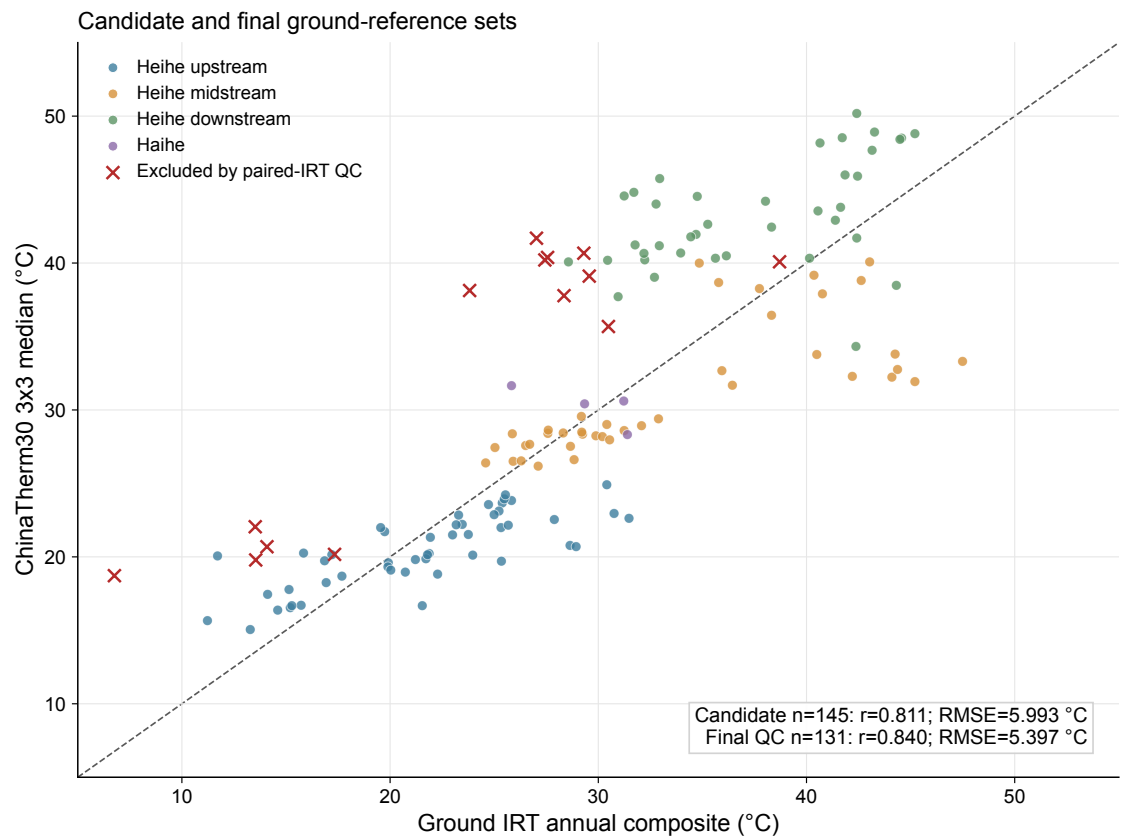


Figure S6: ChinaTherm30 3×3 neighbourhood medians compared with annual ground-IRT composites before and after ground-reference quality control. Coloured circles are the 131 final station-years, and red crosses are the 14 records excluded by the paired-IRT criterion. Summary statistics report the complete 145-record candidate set and the 131-record final set.

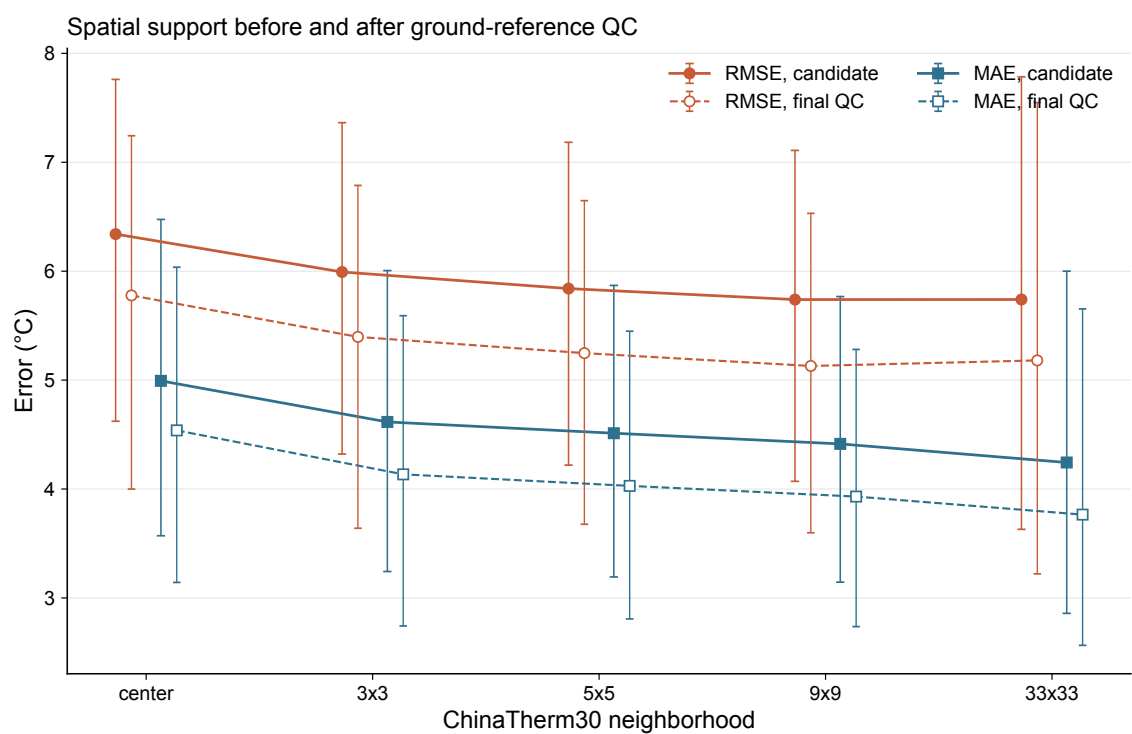


Figure S7: Spatial-support sensitivity before and after ground-reference quality control. Points show RMSE and MAE for the ChinaTherm30 centre pixel and median neighbourhoods from 3×3 to 33×33 pixels; error bars are 95% site-cluster bootstrap intervals.

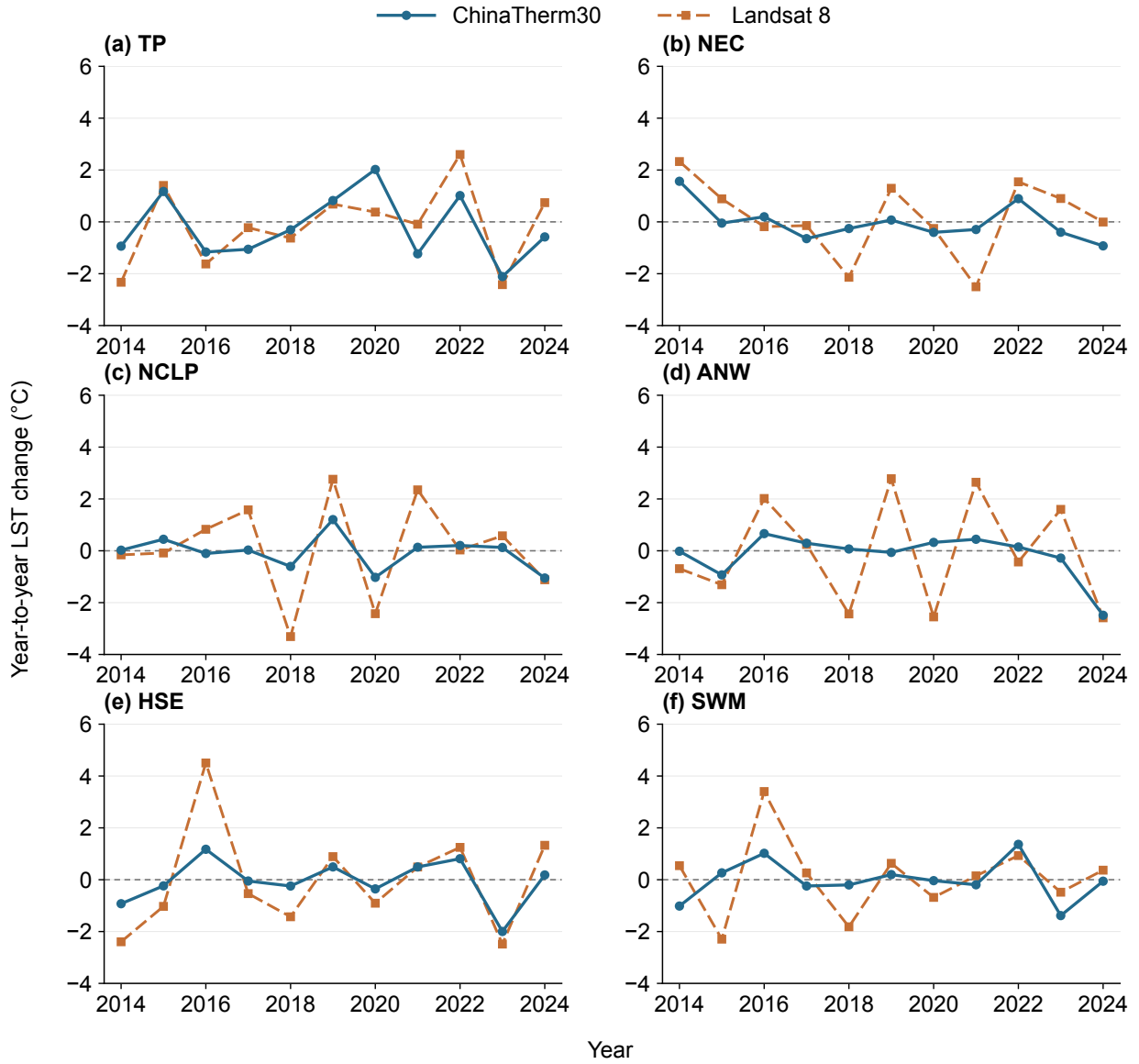


Figure S8: Interannual consistency of regional LST changes. Panels a–f show TP, NEC, NCLP, ANW, HSE and SWM. Curves show area-weighted changes between consecutive annual composites of ChinaTherm30 and Landsat 8; year t denotes the change from $t - 1$ to t . Each macroregion uses the same supported spatial strata across all 11 year pairs during 2013–2024, with fixed stratum-area weights. Both datasets use the same eligible locations within each year pair, while locations may differ among pairs. Horizontal dashed lines indicate zero change.

Supplementary Tables

Table S1: Station metadata and ground-validation coverage. Heihe Site IDs 1–23 follow Liu et al.¹; H1 and H2 denote the Guantao and Daxing sites in the Haihe basin. Candidate records use $K \geq 0.65$, 10:30 local mean solar time ± 30 min, and at least eight valid MOD11A2-aligned eight-day bins; evaluated records additionally satisfy the paired-IRT ground-reference criterion. Observation durations are reproduced from source metadata; evaluated year ranges may contain gaps.

Site ID	Station	Network	Reach	Landscape	Lon.	Lat.	Elev.	Duration reported	Years evaluated	Site-years	Status
1	Jingyangling	Heihe	Up.	alpine meadow	101.120	37.840	3750	Aug. 2013-present	2014-2024	5	Final QC
2	E'bao	Heihe	Up.	alpine meadow	100.920	37.950	3294	June 2013-Oct. 2016	2013-2016	4	Final QC
3	Huangcaogou	Heihe	Up.	alpine meadow	100.730	38.000	3137	June 2013-Apr. 2015	2013-2014	2	Final QC
4	Arou Sunny	Heihe	Up.	alpine meadow	100.520	38.090	3529	Aug. 2013-Aug. 2015	2013-2015	3	Final QC
5	Arou	Heihe	Up.	subalpine meadow	100.460	38.050	3033	June 2008-present	2013-2024	12	Final QC
6	Arou Shady	Heihe	Up.	alpine meadow	100.410	37.980	3536	Aug. 2013-Aug. 2015	2013-2014	2	Final QC
7	Yakou	Heihe	Up.	alpine meadow	100.240	38.010	4148	Jan. 2014-present	2016-2024	9	Final QC
8	Huangzangsi	Heihe	Up.	wheat	100.190	38.230	2612	June 2013-Apr. 2015	2013-2014	2	Final QC
9	Dashalong	Heihe	Up.	alpine marsh	98.940	38.840	3739	Aug. 2013-present	2014-2023	10	Final QC
10	Guantan	Heihe	Up.	Qinghai spruce	100.250	38.530	2835	2008-2011	–	0	No IRT work-book
11	Huazhaizi	Heihe	Mid.	Kalidium desert	100.320	38.770	1731	June 2012-present	2015-2016	2	Final QC
12	Shenshawo	Heihe	Mid.	sandy desert	100.490	38.790	1594	June 2012-Apr. 2015	2012-2014	3	Final QC
13	Heihe RS	Heihe	Mid.	grassland	100.480	38.830	1560	Aug. 2014-present	2016-2024	9	Final QC
14	Zhangye wetland	Heihe	Mid.	reed	100.450	38.980	1460	June 2012-present	2013-2024	11	Final QC
15	Daman	Heihe	Mid.	maize	100.370	38.860	1556	May 2012-present	2013-2024	12	Final QC
16	Yingke	Heihe	Mid.	maize	100.410	38.860	1519	2008-2011	–	0	No IRT work-book
17	Bajitan Gobi	Heihe	Mid.	Reaumuria	100.300	38.920	1562	May 2012-Apr. 2015	2012-2014	3	Final QC
18	Mixed Forest	Heihe	Down.	Populus-Tamarix	101.130	41.990	874	July 2013-present	2013-2024	12	Final QC
19	Barren Land	Heihe	Down.	bare land	101.130	42.000	878	July 2013-Mar. 2016	2015	1	Final QC
20	Sidaoqiao	Heihe	Down.	Tamarix	101.140	42.000	873	July 2013-present	2013-2024	12	Final QC
21	Cropland	Heihe	Down.	melon	101.130	42.000	875	July 2013-Nov. 2015	2013-2014	2	Final QC
22	Populus	Heihe	Down.	Populus	101.120	41.990	876	July 2013-Apr. 2016	2015	1	Final QC
23	Desert	Heihe	Down.	Reaumuria	100.990	42.110	1054	Apr. 2015-present	2015-2024	10	Final QC
H1	Guantao	Haihe	–	–	115.117	36.500	–	2008-2010 dataset scope	2010	1	Final QC
H2	Daxing	Haihe	–	–	116.416	39.616	–	2008-2010 dataset scope	2008-2010	3	Final QC

Table S2: Ground-IRT processing and validation protocol.

Step	Definition
Source screening	Identify station and observation year; retain records with interpretable timestamps and surface IRT measurements, and remove duplicate observations.
Time basis	Interpret station timestamps as China Standard Time and convert to local mean solar time using station longitude.
Nominal overpass	Use 10:30 local mean solar time with a +/-30 min primary window; +/-60 and +/-90 min are sensitivity tests.
IRT consensus	Average IRT-1 and IRT-2 when both are valid; reject paired observations differing by more than 5 degrees C; retain a single valid sensor when only one is available.
Daily aggregation	Take the median of valid observations within each station-day and overpass window.
Clear-sky protocol	Use downward shortwave radiation to calculate a clear-sky index; $K \geq 0.65$ is primary, with all-sky and $K \geq 0.60$ alternatives.
Eight-day alignment	Assign daily values to MOD11A2 nominal eight-day bins whose start DOY is 150-300; average valid daily values within each bin.
Annual composite	Take the median of qualified eight-day means and require at least eight valid bins per station-year.
Ground-reference QC	Apply the paired-IRT consistency criterion to 145 candidate station-years and retain 131 records across all 23 sites. Exclude a station-year when its median daily paired-IRT absolute difference exceeds 5 degrees C; product values and residuals are not used.
Product extraction	Extract the colocated ChinaTherm30 center pixel and median neighborhoods of 3x3, 5x5, 9x9, and 33x33 pixels.
Metrics and uncertainty	Report Pearson r , RMSE, MAE, bias, site-stratified results, and 2,000-replicate site-cluster bootstrap intervals.

The temporal, clear-sky and annual-support filters yielded 145 candidate station-years. Ground-reference quality control excluded a station-year when the median of its daily paired-IRT absolute differences exceeded 5 °C; no ChinaTherm30 or RF-LST value was used in this decision. The final set retained 131 station-years across all 23 sites and excluded 14 records (eight at Huazhaizi Desert Steppe, five at Jingyangling, and one at Barren Land). For the ChinaTherm30 3×3 comparison, r and RMSE changed from 0.811 and 5.993 °C in the candidate set to 0.840 and 5.397 °C in the final set. Table S3 provides the complete exclusion audit.

Table S3: Station-years excluded from the final ground-reference set by paired-IRT inconsistency. Paired days count days with a finite IRT-1/IRT-2 diagnostic; medians and 95th percentiles are daily absolute paired-sensor differences. No satellite-product residual was used in the criterion.

Station	Chinese name	Year	Paired days	Median (°C)	P95 (°C)	Status
Barren Land	裸地	2013	85	5.190	6.098	Excluded
Huazhaizi Desert Steppe	花 寨 子 荒漠	2017	112	5.225	7.202	Excluded
Huazhaizi Desert Steppe	花 寨 子 荒漠	2018	108	8.757	11.329	Excluded
Huazhaizi Desert Steppe	花 寨 子 荒漠	2019	117	8.005	13.871	Excluded
Huazhaizi Desert Steppe	花 寨 子 荒漠	2020	122	8.830	14.162	Excluded

Table S3 continued

Station	Chinese name	Year	Paired days	Median (°C)	P95 (°C)	Status
Huazhaizi Desert Steppe	花 寨 子 荒漠	2021	109	10.620	15.492	Excluded
Huazhaizi Desert Steppe	花 寨 子 荒漠	2022	89	10.520	13.713	Excluded
Huazhaizi Desert Steppe	花 寨 子 荒漠	2023	120	9.425	13.892	Excluded
Huazhaizi Desert Steppe	花 寨 子 荒漠	2024	115	8.150	12.877	Excluded
Jingyangling	景阳岭	2018	107	6.305	10.326	Excluded
Jingyangling	景阳岭	2019	115	7.570	16.906	Excluded
Jingyangling	景阳岭	2020	110	8.273	12.861	Excluded
Jingyangling	景阳岭	2022	106	5.447	9.921	Excluded
Jingyangling	景阳岭	2023	121	6.960	9.720	Excluded

Regional interannual consistency

Sampling and temporal pairing

The analysis compares consecutive-year changes in ChinaTherm30 and Landsat 8 annual LST composites during 2013–2024. The Landsat date window, temperature conversion and strict quality screening follow the main-text Landsat 8 comparison. Each year requires at least three valid observations; both annual ChinaTherm30 3×3 means must be available, and neighbourhood water fractions must not exceed 10%. The original 20,000 candidate points were supplemented by a pool of 59,626 candidates to improve representation where cloud screening had reduced temporal support. Within each 4° tile–macroregion–elevation stratum, eligible original points were retained first, followed by eligible reserve points in their preassigned order until the original stratum quota was reached.

For each of the 11 adjacent-year pairs, both datasets use exactly the same selected locations. A stratum is included only if it has at least three paired points and a Kish effective sample size of at least three in every year pair. The retained stratum set and its area weights remain fixed across all pairs; individual points may change between pairs. Original 2° sampling weights are normalized within each retained stratum to its original area. Supported area fractions are relative to the full macroregional sampling-design area, not the fraction of pixels directly observed.

Metrics and uncertainty

For product P , the regional change ending in year t is

$$\Delta T_{r,t}^P = \frac{\sum_{s \in S_r} A_s \left[\sum_{i \in I_{s,t}} w_i (T_{i,t}^P - T_{i,t-1}^P) / \sum_{i \in I_{s,t}} w_i \right]}{\sum_{s \in S_r} A_s}, \quad (1)$$

where S_r is the fixed set of supported strata in region r , A_s is stratum area, and $I_{s,t}$ contains the paired locations. Pearson correlation and RMSE are calculated from the 11 regional change values for the two datasets. The standard-deviation ratio is $\text{SD}(\Delta T^{\text{CT30}}) / \text{SD}(\Delta T^{\text{L8}})$; it describes variability in adjacent-year changes, not the variability or trend of annual temperature levels. Direction agreement counts year pairs for which both changes have the same sign.

Confidence intervals use 1,000 spatial-cluster bootstrap replicates over 4° tiles. They are conditional on the observed years and supported domain and do not quantify uncertainty over unobserved regions or resampled years. SWM contains only nine spatial blocks. Observation-date summaries first calculate the median valid-observation DOY for each point and year, then take an area-weighted median across paired points; they are not the first and last acquisition dates of a season.

Results

Regional change correlations range from 0.399 in SWM to 0.870 in HSE, and change RMSE ranges from 1.017 to 1.752 $^\circ\text{C}$ (Figure S8; Tables S4–S5). HSE agrees in change direction for all 11 pairs and TP for ten pairs; ANW and SWM agree for six. Standard-deviation ratios of 0.349–0.826 indicate smaller adjacent-year changes in ChinaTherm30 than in Landsat 8 across all six regions. Fixed strata represent more than 99% of the design area in TP, NEC, NCLP and ANW, compared with 83.3% in HSE and 62.5% in SWM. Year-pair sample sizes, temperature changes and observation support are reported in Table S6.

Table S4: Regional interannual-change metrics for 2013–2024 (11 year pairs). Supported area denotes the fixed-stratum fraction of the design area. SD ratio is ChinaTherm30/Landsat 8.

Region	Area (%)	Paired points	r	RMSE ($^\circ\text{C}$)	SD ratio	Same sign
TP	99.2	3,841–4,228	0.734	1.017	0.826	10/11
NEC	99.9	3,689–3,691	0.592	1.158	0.481	8/11
NCLP	99.3	1,871–1,883	0.773	1.363	0.349	8/11
ANW	99.6	4,381–4,417	0.465	1.752	0.423	6/11
HSE	83.3	1,665–2,334	0.870	1.269	0.432	11/11

Region	Area (%)	Paired points	r	RMSE (°C)	SD ratio	Same sign
SWM	62.5	407–674	0.399	1.329	0.520	6/11

Table S5: Spatial-cluster bootstrap 95% intervals for the metrics in Table S4 (1,000 replicates).

Region	Tile blocks	r	RMSE (°C)	SD ratio
TP	28 [0.394, 0.874]	[0.857, 1.344]	[0.706, 1.113]	
NEC	24 [0.213, 0.764]	[0.790, 2.000]	[0.311, 0.673]	
NCLP	14 [0.496, 0.858]	[1.018, 2.062]	[0.272, 0.494]	
ANW	27 [0.200, 0.689]	[1.445, 2.159]	[0.360, 0.529]	
HSE	15 [0.759, 0.905]	[0.836, 1.904]	[0.325, 0.613]	
SWM	9 [0.158, 0.475]	[1.226, 1.839]	[0.318, 0.655]	

Table S6: Paired-year regional changes and observation support. Temperature changes are in degrees Celsius. Count and DOY columns give area-weighted medians for the earlier/later year, respectively; DOY summarizes point-level median observation dates.

Region	Year pair	Points	ΔT_{CT30}	ΔT_{L8}	Count	DOY
TP	2013–2014	4,080	-0.938	-2.330	5/5	224/238
TP	2014–2015	4,113	1.173	1.405	5/6	238/242
TP	2015–2016	4,165	-1.161	-1.624	6/5	242/238
TP	2016–2017	4,025	-1.057	-0.224	5/5	239/248
TP	2017–2018	3,841	-0.305	-0.627	5/5	247/239
TP	2018–2019	3,843	0.823	0.685	5/5	237/237
TP	2019–2020	4,022	2.021	0.377	5/6	237/250
TP	2020–2021	4,224	-1.232	-0.091	6/5	251/242
TP	2021–2022	4,158	1.014	2.598	6/5	242/221
TP	2022–2023	4,121	-2.110	-2.422	5/5	222/242
TP	2023–2024	4,228	-0.581	0.741	5/6	242/235
NEC	2013–2014	3,691	1.566	2.327	6/7	246/238
NEC	2014–2015	3,691	-0.048	0.889	7/8	238/233
NEC	2015–2016	3,691	0.196	-0.185	8/7	233/234
NEC	2016–2017	3,691	-0.647	-0.144	7/8	234/234
NEC	2017–2018	3,691	-0.257	-2.139	8/7	236/246
NEC	2018–2019	3,691	0.071	1.295	7/7	246/250
NEC	2019–2020	3,691	-0.405	-0.255	7/6	250/232
NEC	2020–2021	3,691	-0.295	-2.504	7/6	232/249
NEC	2021–2022	3,689	0.895	1.545	6/7	250/248
NEC	2022–2023	3,689	-0.400	0.901	7/8	248/249
NEC	2023–2024	3,691	-0.926	-0.007	8/7	249/242
NCLP	2013–2014	1,878	0.024	-0.158	6/5	235/232
NCLP	2014–2015	1,877	0.445	-0.085	6/7	232/239
NCLP	2015–2016	1,877	-0.104	0.829	7/5	238/239
NCLP	2016–2017	1,871	0.029	1.579	5/5	238/216
NCLP	2017–2018	1,874	-0.604	-3.310	5/6	217/248
NCLP	2018–2019	1,883	1.203	2.760	6/6	249/229
NCLP	2019–2020	1,878	-1.025	-2.427	6/5	229/241
NCLP	2020–2021	1,877	0.135	2.351	5/6	241/227
NCLP	2021–2022	1,882	0.204	0.031	5/7	226/223
NCLP	2022–2023	1,883	0.128	0.577	6/6	224/235
NCLP	2023–2024	1,882	-1.050	-1.118	6/6	234/232
ANW	2013–2014	4,417	-0.018	-0.689	8/9	227/228
ANW	2014–2015	4,416	-0.930	-1.310	9/9	228/230
ANW	2015–2016	4,412	0.662	2.018	9/8	230/226
ANW	2016–2017	4,408	0.293	0.245	9/9	226/228
ANW	2017–2018	4,397	0.069	-2.436	9/9	228/236

Region	Year pair	Points	ΔT_{CT30}	ΔT_{L8}	Count	DOY
ANW	2018–2019	4,392	-0.063	2.781	9/9	236/224
ANW	2019–2020	4,398	0.324	-2.550	9/9	224/233
ANW	2020–2021	4,417	0.445	2.644	9/9	233/228
ANW	2021–2022	4,410	0.147	-0.431	9/9	228/231
ANW	2022–2023	4,396	-0.281	1.598	9/9	231/227
ANW	2023–2024	4,381	-2.488	-2.585	9/8	227/224
HSE	2013–2014	2,334	-0.926	-2.395	5/4	239/247
HSE	2014–2015	2,105	-0.238	-1.028	4/4	248/257
HSE	2015–2016	2,015	1.174	4.504	4/4	259/227
HSE	2016–2017	1,785	-0.051	-0.537	4/4	223/230
HSE	2017–2018	1,808	-0.244	-1.430	4/4	230/233
HSE	2018–2019	2,094	0.493	0.890	4/4	233/246
HSE	2019–2020	1,840	-0.354	-0.906	5/4	247/239
HSE	2020–2021	1,881	0.493	0.491	4/4	241/251
HSE	2021–2022	2,231	0.810	1.247	4/5	251/245
HSE	2022–2023	1,868	-1.997	-2.483	5/4	245/239
HSE	2023–2024	1,665	0.183	1.327	4/4	239/238
SWM	2013–2014	648	-1.016	0.542	5/4	238/221
SWM	2014–2015	547	0.265	-2.293	4/4	218/235
SWM	2015–2016	534	1.019	3.402	4/4	239/217
SWM	2016–2017	407	-0.242	0.262	4/4	215/223
SWM	2017–2018	443	-0.201	-1.817	4/4	219/227
SWM	2018–2019	624	0.198	0.632	4/4	227/225
SWM	2019–2020	474	-0.037	-0.683	4/4	227/214
SWM	2020–2021	448	-0.196	0.148	4/4	214/239
SWM	2021–2022	582	1.364	0.932	4/4	227/228
SWM	2022–2023	674	-1.384	-0.482	4/4	226/225
SWM	2023–2024	670	-0.057	0.368	4/4	225/232

References

- [1] Liu, S. et al. The Heihe Integrated Observatory Network: A basin-scale land surface processes observatory in China. In: *Vadose Zone Journal* **17.1** (2018), p. 180072. <https://doi.org/10.2136/vzj2018.04.0072>.